

Rithani Saravanakumar

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EDUCATION

National University of Singapore

Expected Graduation: May 2027

Bachelor of Computing (Honours) in Computer Science

Relevant Coursework: Foundations of Machine Learning, Big Data Systems for Data Science, Data Structures and Algorithms, Design and Analysis of Algorithms, Database Systems, Operating Systems, Software Engineering

SKILLS

Generative AI: RAG, Agentic AI, Multi-Agent Systems, Embeddings, Prompt Engineering, System Prompts

Evaluation & Safety: LLM Evals, Agent Tracing, Regression Testing, Ablations, Guardrails, Human-in-the-Loop

Machine Learning: Model Training & Evaluation, Multimodal ML, Computer Vision, NLP, Time-Series Modeling

Languages & Frameworks: Python, TypeScript, Go, Java, SQL, PyTorch, TensorFlow, scikit-learn, LangChain

Cloud, Data & Tools: AWS, GCP, Azure, Docker, Git, Pinecone, Weaviate, PostgreSQL, MongoDB, BigQuery

EXPERIENCE

AI/Software Engineer Intern | *Python, React.js, Next.js, Postgres, Docker, Azure, Agno*

Aug 2025 – Jul 2026

NiaHealth (Health Tech Startup, Series A)

Toronto, Ontario

- Prevented **8 report errors** before patient delivery with an Agno validation agent that checks reports against clinical recommendations, keeps PII out of its inputs, and requires clinician sign-off.
- Improved food-photo macro extraction accuracy **27%** over a GPT vision baseline by adding nutrition-database retrieval to an Agno multi-agent pipeline, evaluated on Google Nutrition5k.
- Cut medical-report processing **70% (10 to 3 mins)** across **200+ reports/week**, saving **9 hours/week** of review, with an async GPT pipeline on Azure queue workers with retries, demoed to the COO and CTO.
- Launched a cycle-tracking MVP in React and Python to **8,000+ users** in **12 weeks** as sole engineer, turning clinical interviews into requirements and shipping via unit tests, code review, and CI/CD.
- Scaled ops capacity **9× (8 to 70+ orders/hour)** by designing the PostgreSQL data model and a full-stack CGM dashboard that automates communications and centralizes order tracking.

Machine Learning Researcher, ICU Mortality Prediction | *PyTorch, Multimodal ML*

Aug 2026 – Present

National University of Singapore (Supervisor: Dr. Kabir)

Singapore

- Designing a PyTorch model fusing chest X-ray, ECG, and Neural ODE lab encoders across **3 ICU modalities**.
- Patched a data-leakage bug that mixed features across ICU stays while reproducing a MIMIC-IV AKI baseline.

Software Engineer Intern | *Go, React, Next.js, MongoDB, Pinecone, LangChain*

May 2024 – Aug 2024

Buymed (B2B Pharma Marketplace Startup, Series B)

Ho Chi Minh City, Vietnam

- Cut customer information-retrieval time on the company website from **5 mins to 10 seconds** by launching a RAG chatbot with Pinecone, LangChain, and the OpenAI API.
- Automated nightly pricing collection across **6 pharmaceutical sources** by building a concurrent Go pipeline, **12** authenticated REST API endpoints, and a React dashboard used by a **35-person team**.

PROJECTS

Intent Ledger Shopping Copilot (Team Project) | *Python, SQLite FTS5*

Aug 2026 – Sep 2026

- Scored **8.2x** the baseline, **96.5% Hit Rate@10**, and **0.77 MRR** over **50,000 products** in a **200-session** evaluation by building hybrid retrieval with rank fusion, multi-turn intent state, and fallbacks.

Bharatanatyam Mudra Practice Tool | *Python, MediaPipe, Flask*

May 2026 – Sep 2026

- Improved independent practice **22%** for **20+ students** over **8 weeks** with a live-camera Flask tool whose MediaPipe model hits **88.7% accuracy** on **28 mudras (1,400+ samples)**.

Agent Observability Dashboard | *React, TypeScript, WebSockets, LLM Evals*

Mar 2026 – May 2026

- Built real-time WebSocket tracing of agent tool calls, latency, tokens, and failures, plus an eval harness that tracks failure rates to catch regressions across prompt and model versions.

LEADERSHIP

Teaching Assistant, Digital Ethics and Data Privacy

Aug 2024 – Present

- Facilitated weekly tutorials for **100+ students** on responsible AI, data privacy, algorithmic bias, and governance, guiding discussions of real-world technology ethics cases and bias-mitigation trade-offs.